GENX: Porewater Methane

March and April 2026 Samples

2026-06-11

Contents

##Run Information

```r
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

```
## Run Information: Melanie Giessner
```

```r
#set the run date & user name
sample_year <- "2026"
sample_month <- "March & April"
user <- "Melanie Giessner"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

#change dates
folder_path_rawdata <- paste0("Raw Data")
GHG_files <-  list.files(path = folder_path_rawdata,pattern = "20260406" ,full.names = TRUE)
head(GHG_files)
```

```
## [1] "Raw Data/20260406_GENX_GHGdata.csv"
```

```r
#GHG_files <- c("Raw Data/20260417_GENX_GHGdata.csv","Raw Data/20260501_GENX_GHGdata.csv" )


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/ShimadzuGC_Slope_QAQC.csv"


# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_20260406.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""

#Metadata File Path
  Raw_Metadata = here(".." , "GCReW", "GCReW_Project_Treatment_Metadata.csv")

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

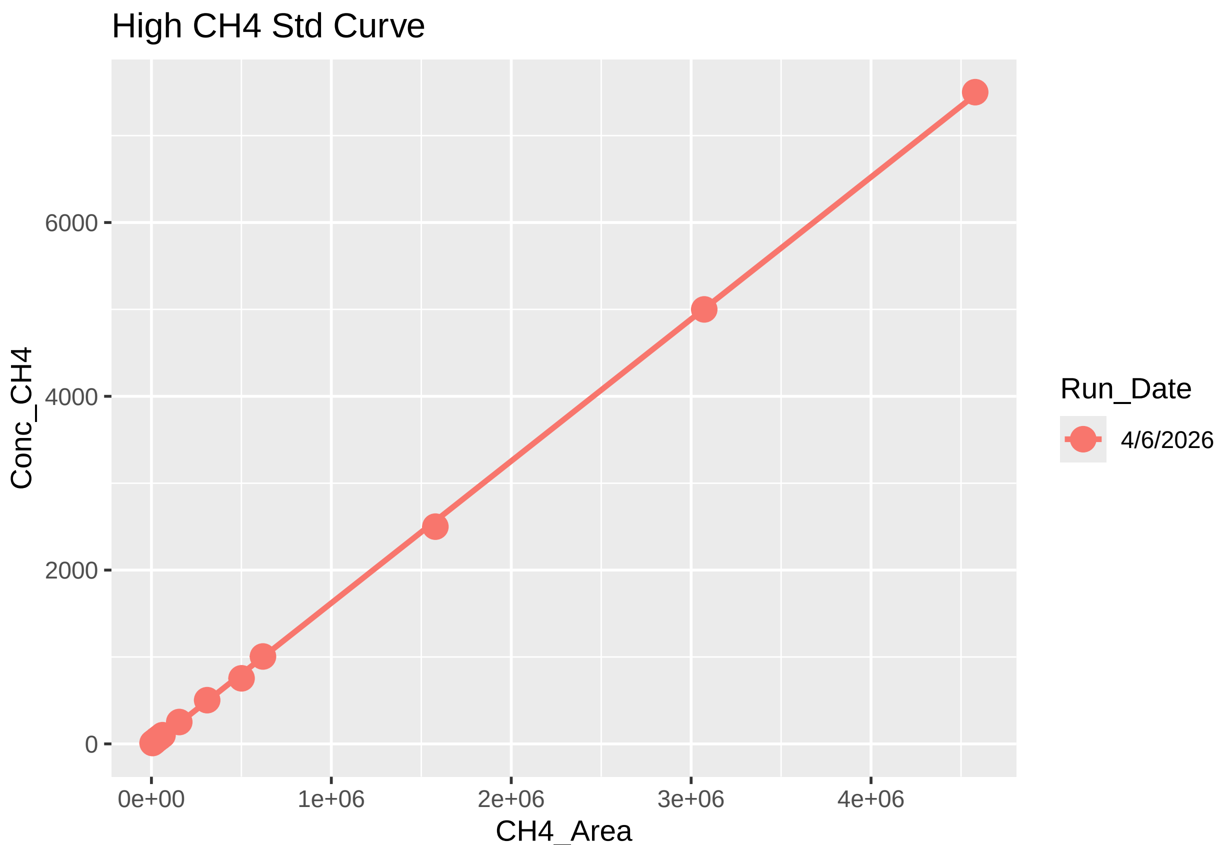
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

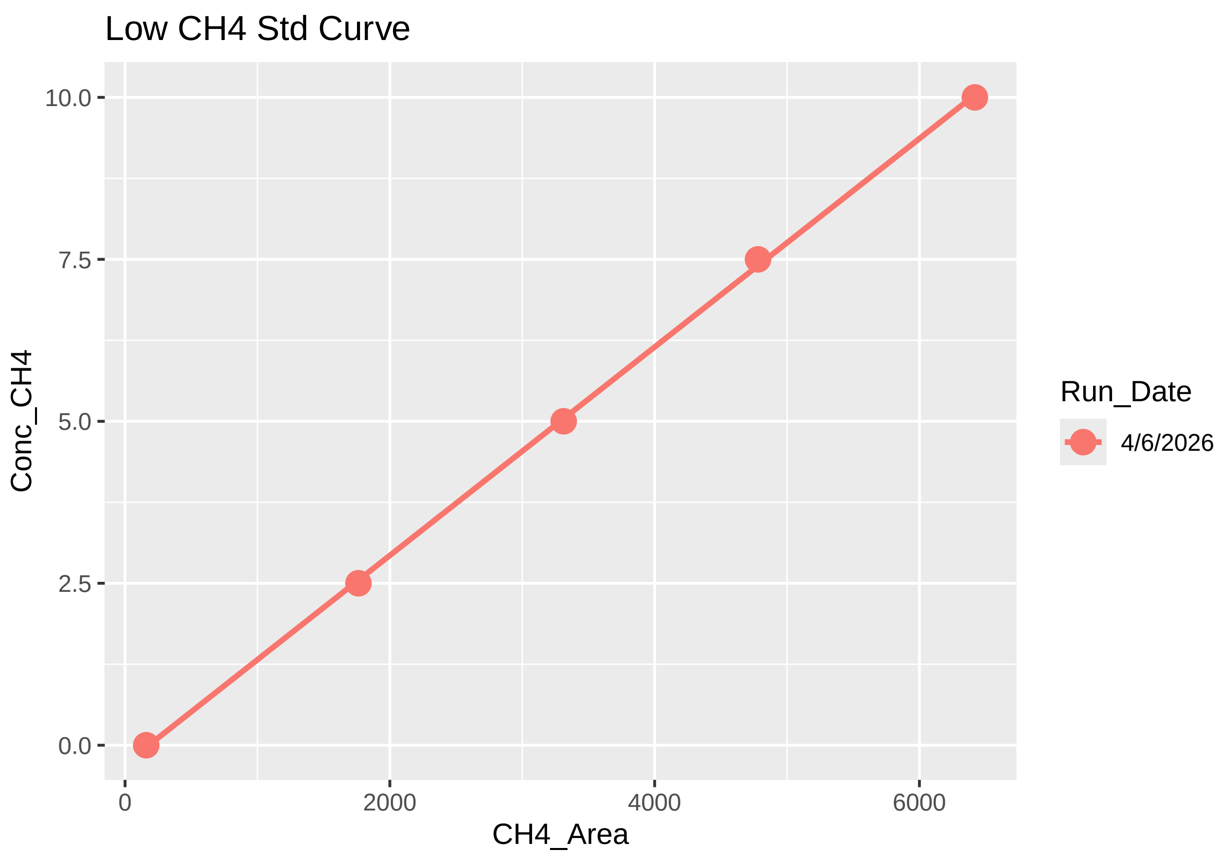
1 Read in data files

2 Assess standard curves

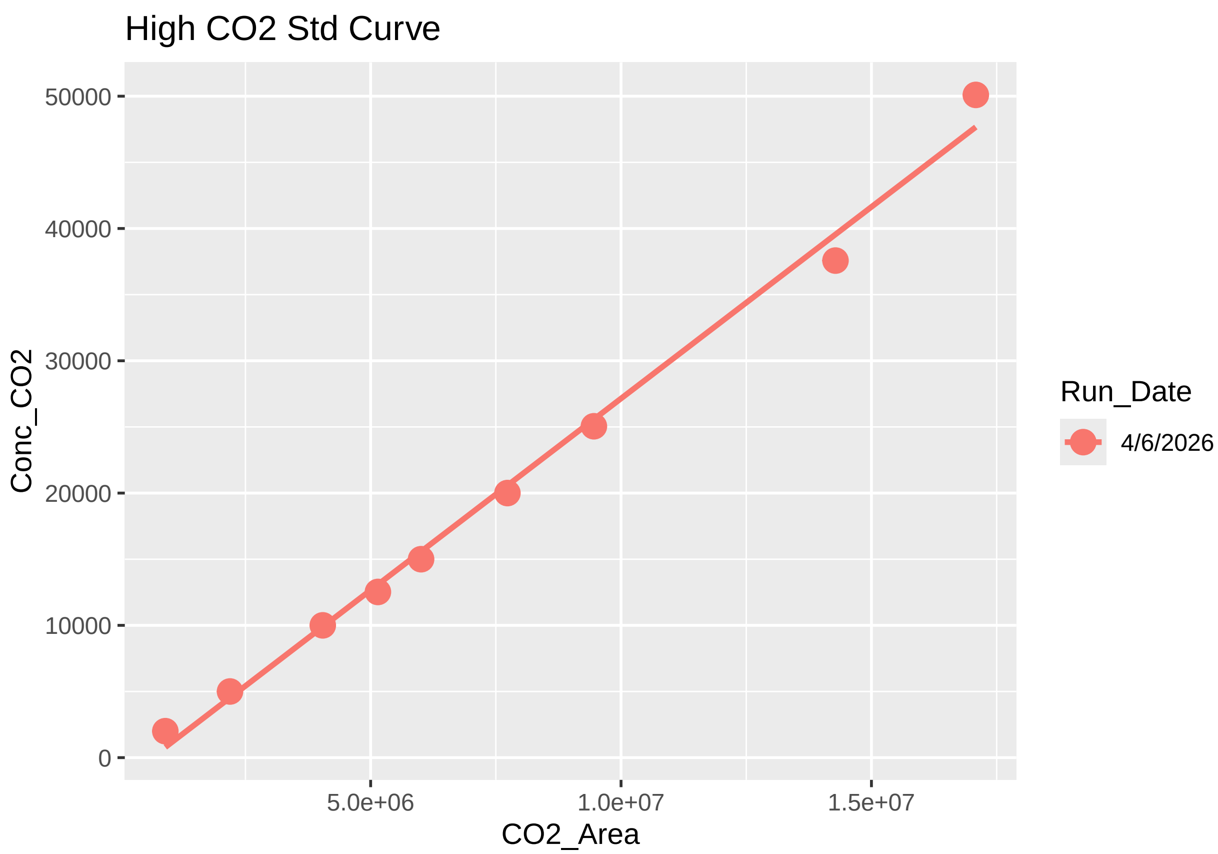
```
## `geom_smooth()` using formula = 'y ~ x'
```



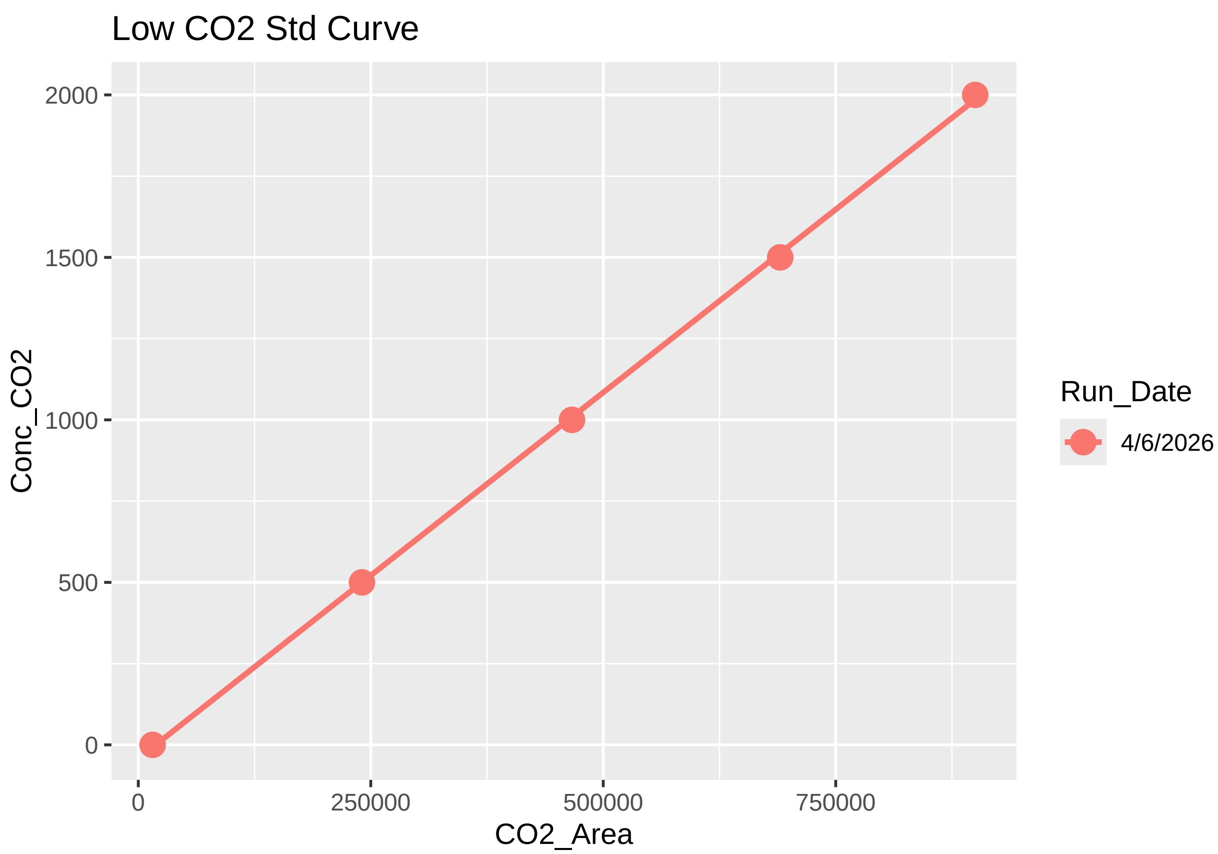

```
## `geom_smooth()` using formula = 'y ~ x'
```



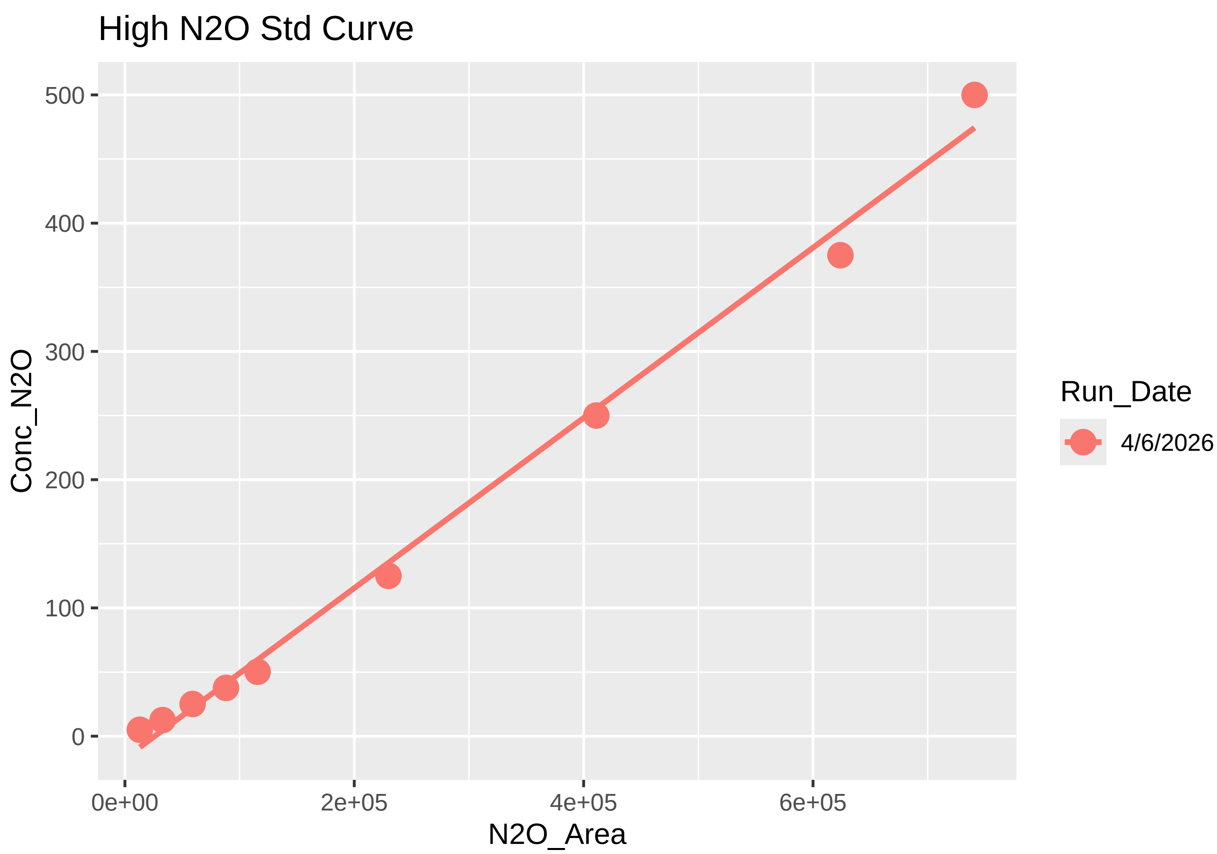

```
## Joining with `by = join_by(Run_Date)`

## [1] "High_R2 is Good-4/6/2026"

## [1] "Low_R2 is Good-4/6/2026"

## `geom_smooth()` using formula = 'y ~ x'
```



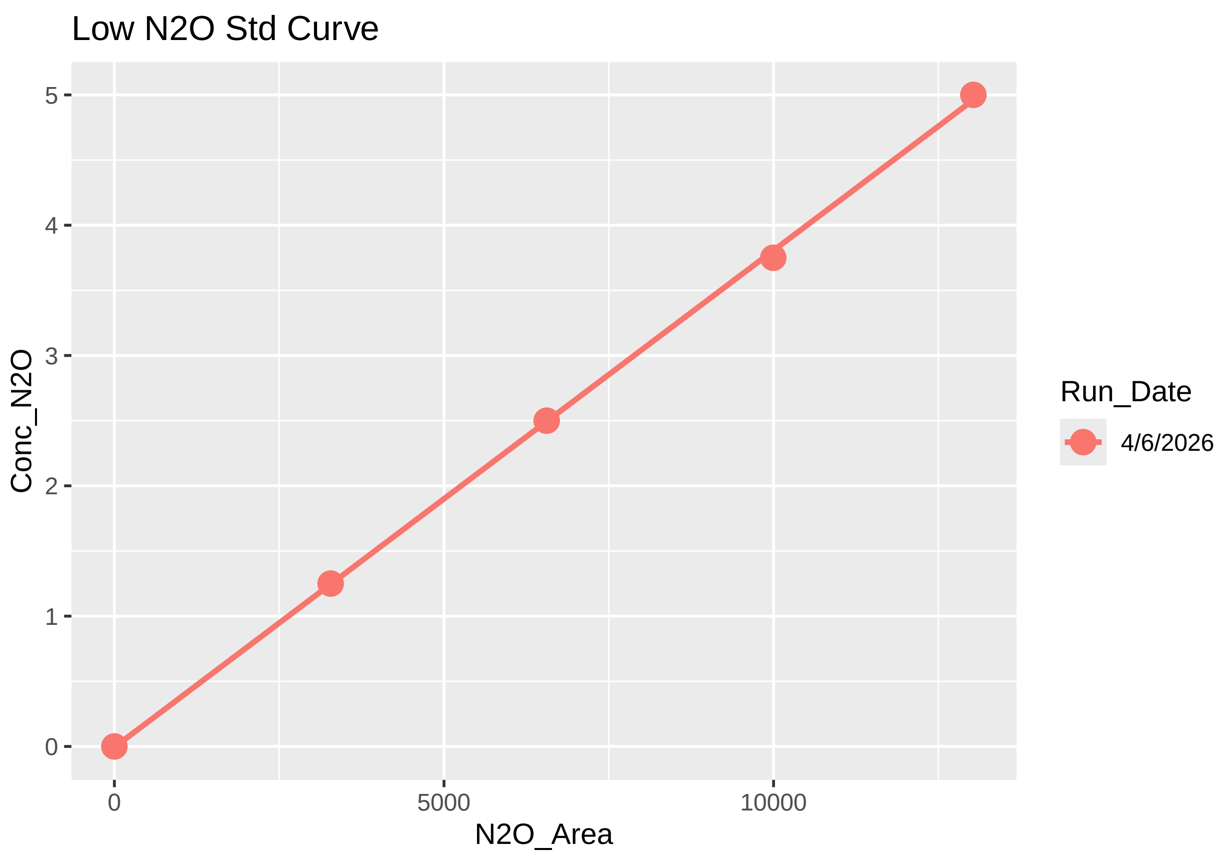

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## Joining with `by = join_by(Run_Date)`

## [1] "High_R2 is Good-4/6/2026"

## [1] "Low_R2 is Good-4/6/2026"

## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## Joining with `by = join_by(Run_Date)`
```

```
## [1] "High_R2 is Good-4/6/2026"
```

```
## [1] "Low_R2 is Good-4/6/2026"
```

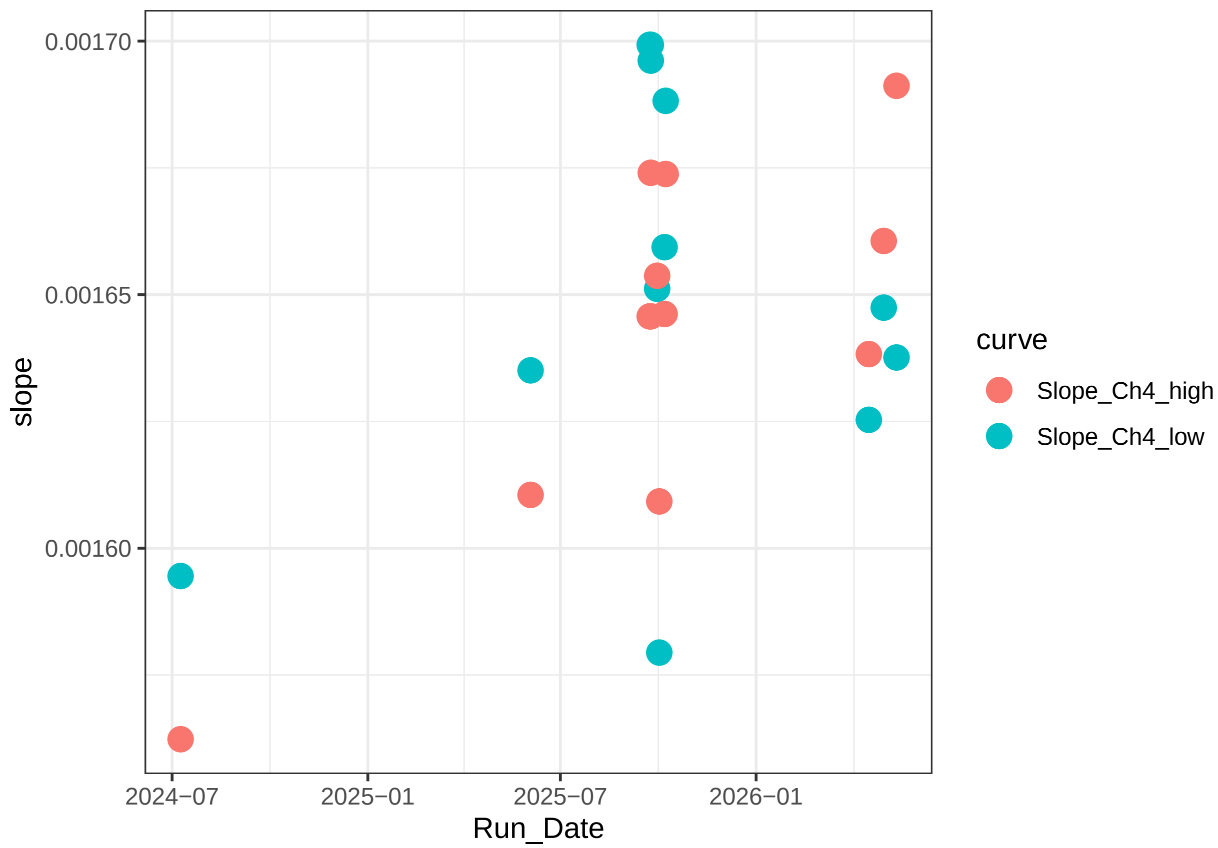
```
#add new standard curve data to log
Slope_CH4_1 <- Slope_CH4 %>% mutate(curve = paste0("Slope_CH4_", CH4_Curve), R2 = CH4_R2, slope = CH4_Sl
  CH4_Curve == "Low" ~ paste(CH4_low_std_crv_cutoff,"ppm"),
  CH4_Curve == "High" ~ paste(CH4_high_std_crv_cutoff,"ppm"),
  .default = as.character(CH4_mid_std_crv_cutoff)
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
Slope_CO2_1 <- Slope_CO2 %>% mutate(curve = paste0("Slope_CO2_", CO2_Curve), R2 = CO2_R2, slope = CO2_Sl
  CO2_Curve == "Low" ~ paste(CO2_low_std_crv_cutoff,"ppm"),
  CO2_Curve == "High" ~ paste(CO2_high_std_crv_cutoff,"ppm"),
  .default = as.character(CO2_mid_std_crv_cutoff)
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
Slope_N2O_1 <- Slope_N2O %>% mutate(curve = paste0("Slope_N2O_", N2O_Curve), R2 = N2O_R2, slope = N2O_Sl
  N2O_Curve == "Low" ~ paste(N2O_low_std_crv_cutoff,"ppm"),
  N2O_Curve == "High" ~ paste(N2O_high_std_crv_cutoff,"ppm"),
  .default = as.character(N2O_mid_std_crv_cutoff)
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)

Slope <- rbind(Slope_CH4_1, Slope_CO2_1, Slope_N2O_1)
```

```r
log <- read.csv(log_path)
log <- log[, -c(1)]
head(log)
```

```
##   Analysis   Run_Date      Run_by Curve_top          curve       slope
## 1      CH4 2024-07-09 Alia Al-Haj    10 ppm  Slope_Ch4_low 0.001594508
## 2      CH4 2024-07-09 Alia Al-Haj   100 ppm Slope_Ch4_high 0.001562326
## 3      N2O 2024-07-09 Alia Al-Haj     5 ppm  Slope_N2O_low 0.000192023
## 4      N2O 2024-07-09 Alia Al-Haj  50.2 ppm Slope_N2O_high 0.000282245
## 5      CO2 2024-07-09 Alia Al-Haj  2000 ppm  Slope_CO2_low 0.000282245
## 6      CO2 2024-07-09 Alia Al-Haj 20000 ppm Slope_CO2_high 0.001549502
##     intercept        R2
## 1 -0.20901179 0.9960215
## 2  0.09646243 0.9996531
## 3  0.07848462 0.9940785
## 4 -0.49925490 0.9960644
## 5 -0.49925490        NA
## 6 38.64567509 0.9996907
```

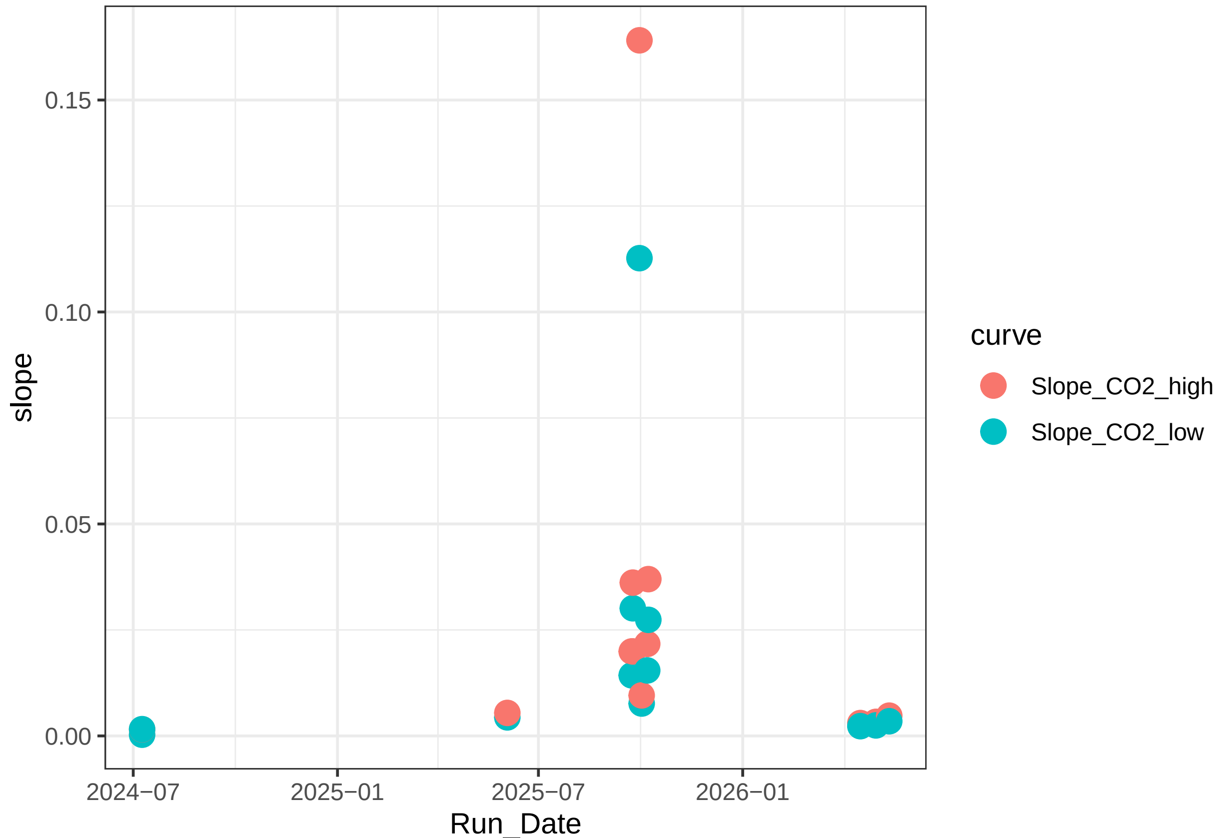
```r
log <- rbind(log, Slope)

#remove possible duplicates from multiple QAQC attempts

unique_log <- log %>%
     mutate(Run_Date = as.Date(Run_Date,tryFormats = c("%Y-%m-%d", "%m/%d/%Y", "%m/%d/%y")), curve= gsu

Slopes_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

Slopes_chk_CH4
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



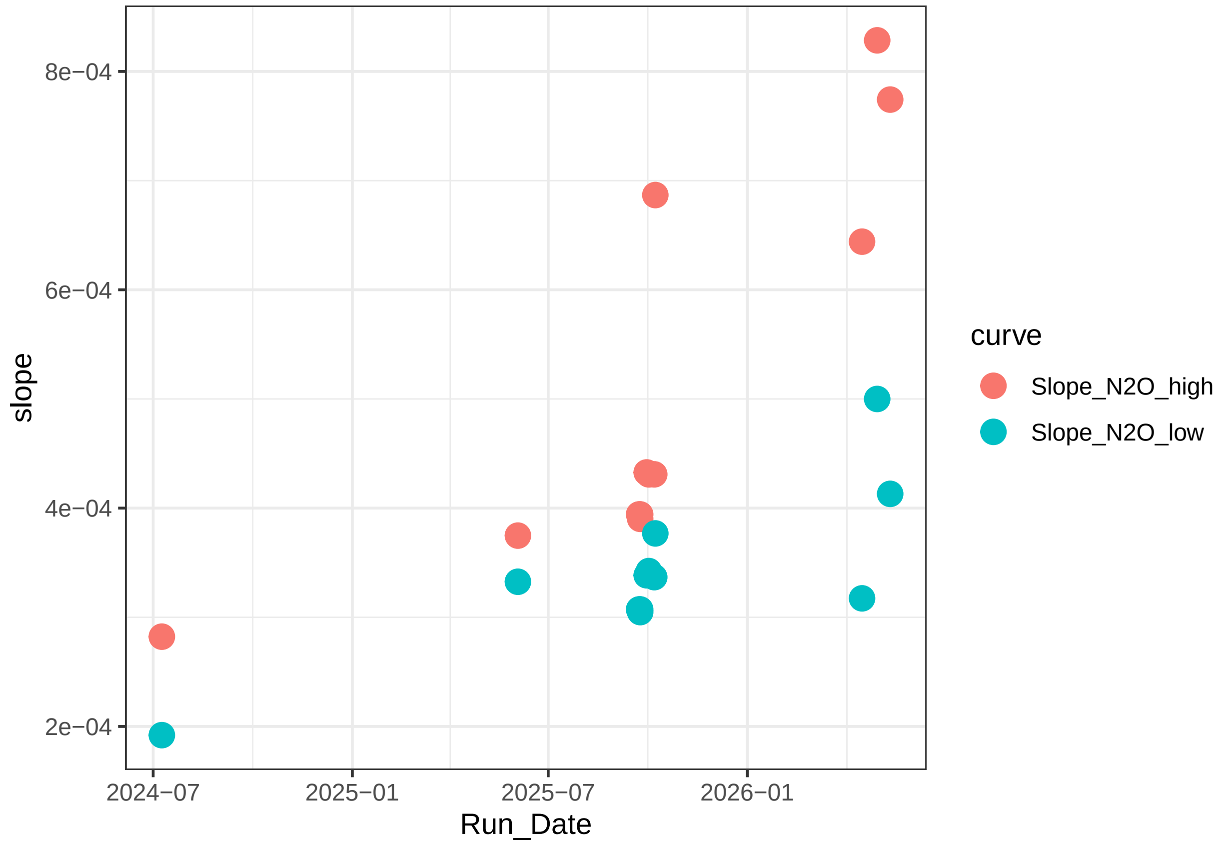

```
Slopes_chk_CO2 <- ggplot(unique_log %>% filter( grepl("CO2",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

Slopes_chk_CO2
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



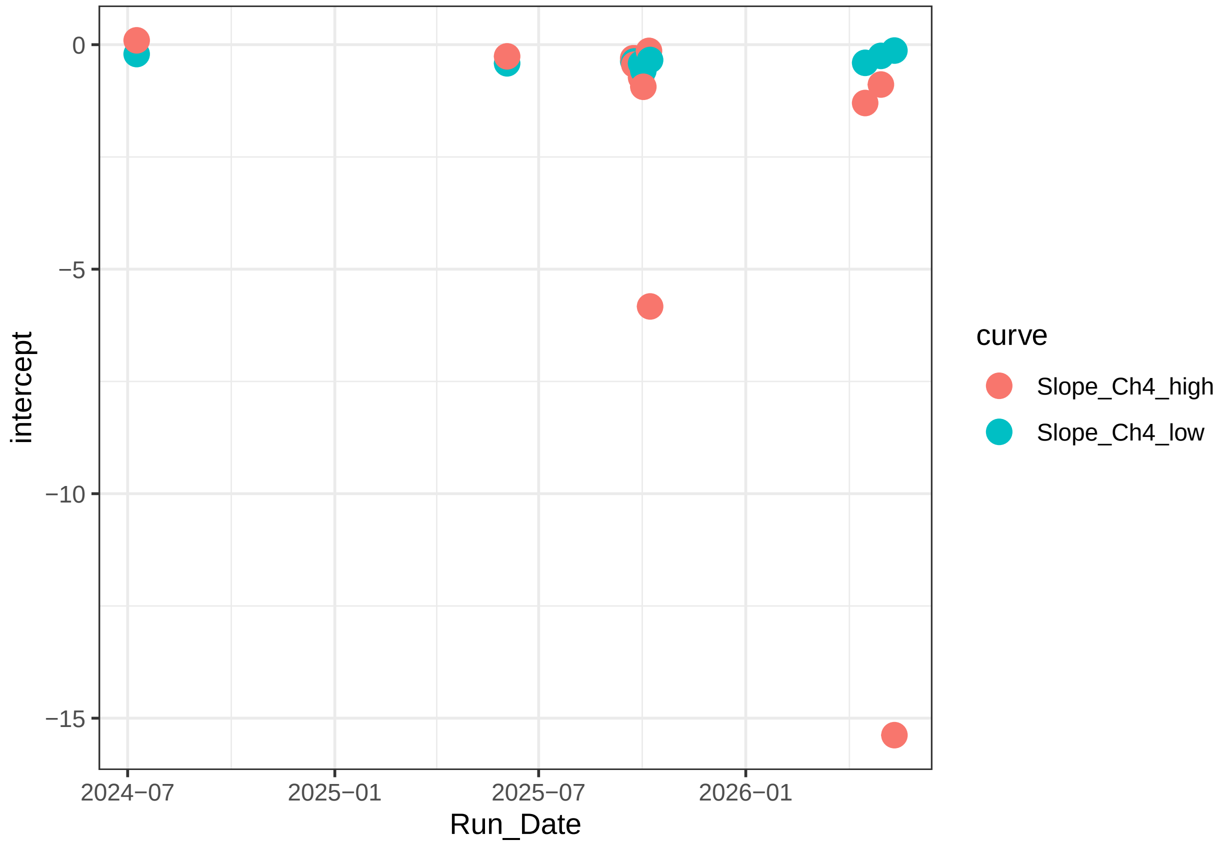

```r
Slopes_chk_N2O <- ggplot(unique_log %>% filter( grepl("N2O",curve,ignore.case = T)), aes(x = Run_Date
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

Slopes_chk_N2O
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



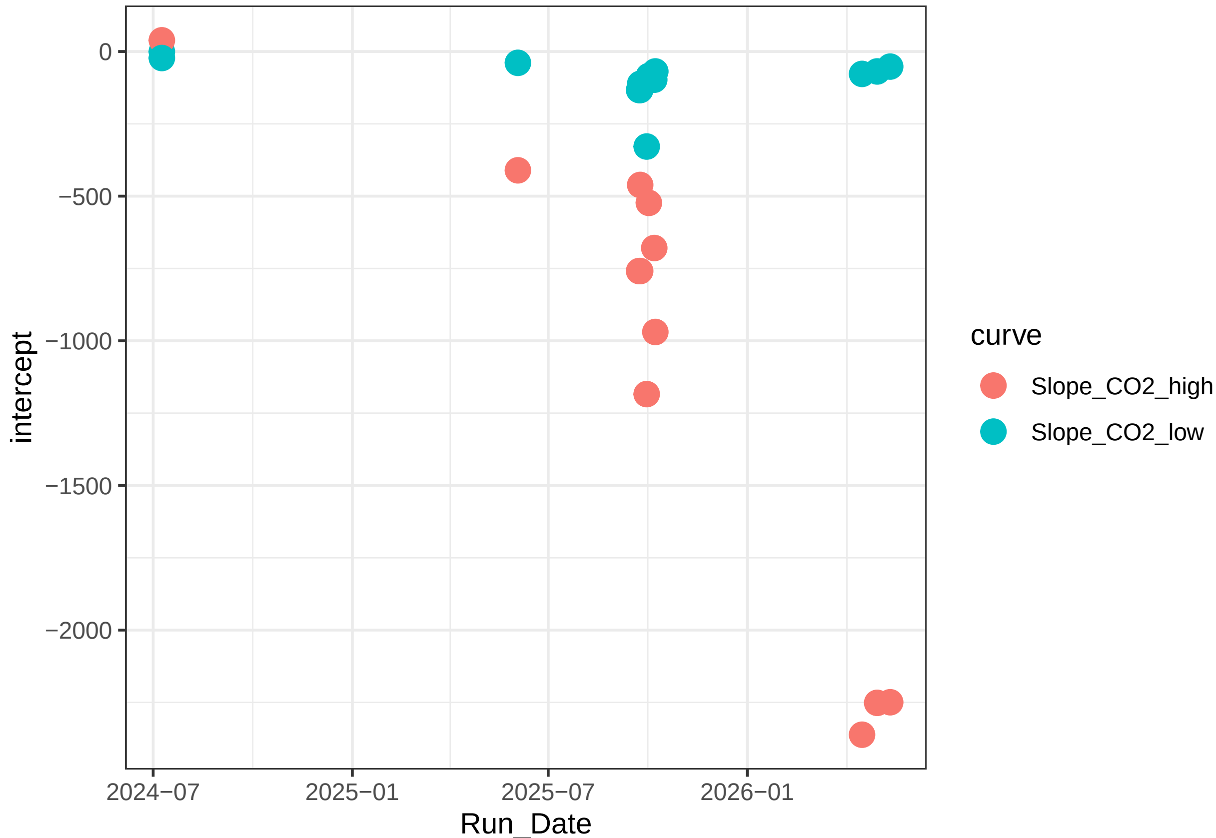

```
int_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date, y
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

int_chk_CH4
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



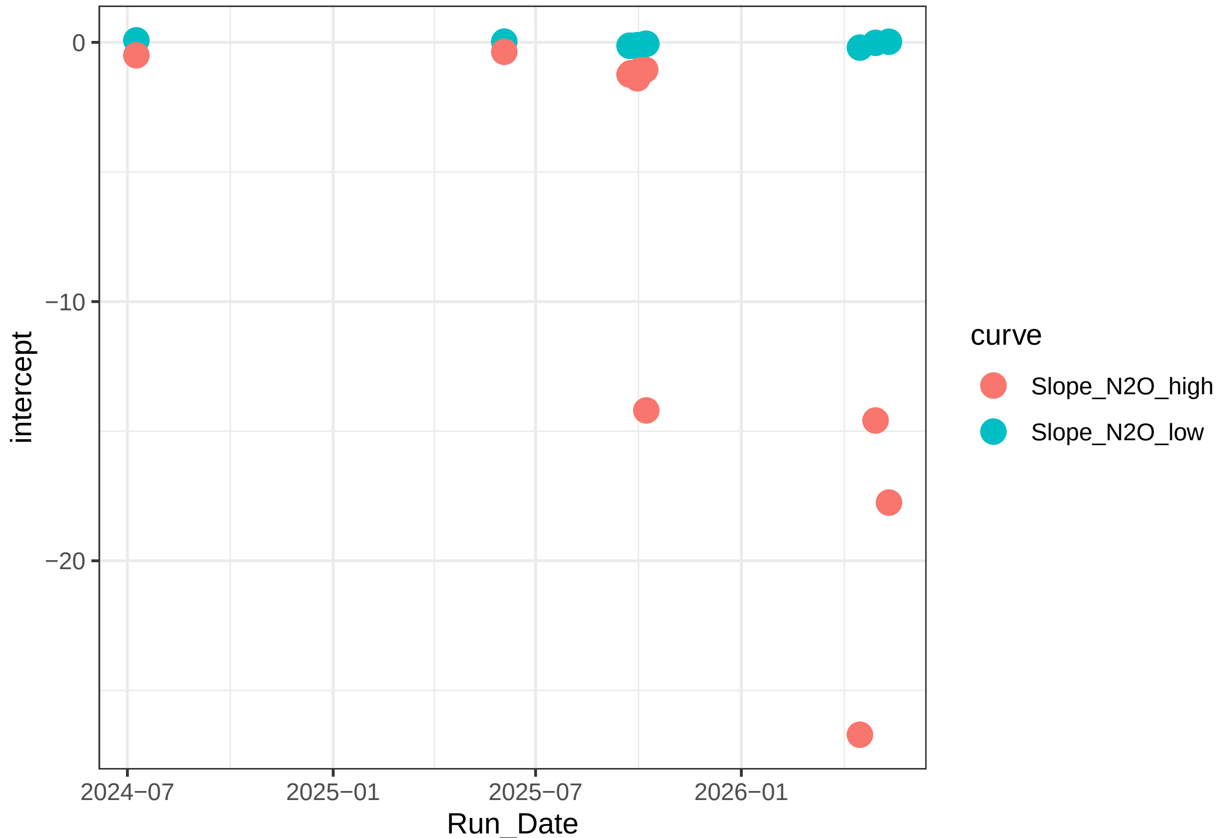

```
  int_chk_CO2 <- ggplot(unique_log %>% filter( grepl("CO2",curve,ignore.case = T)), aes(x = Run_Date, y
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

int_chk_CO2
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```




```
int_chk_N2O <- ggplot(unique_log %>% filter( grepl("N2O",curve,ignore.case = T)), aes(x = Run_Date, y
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

int_chk_N2O
```

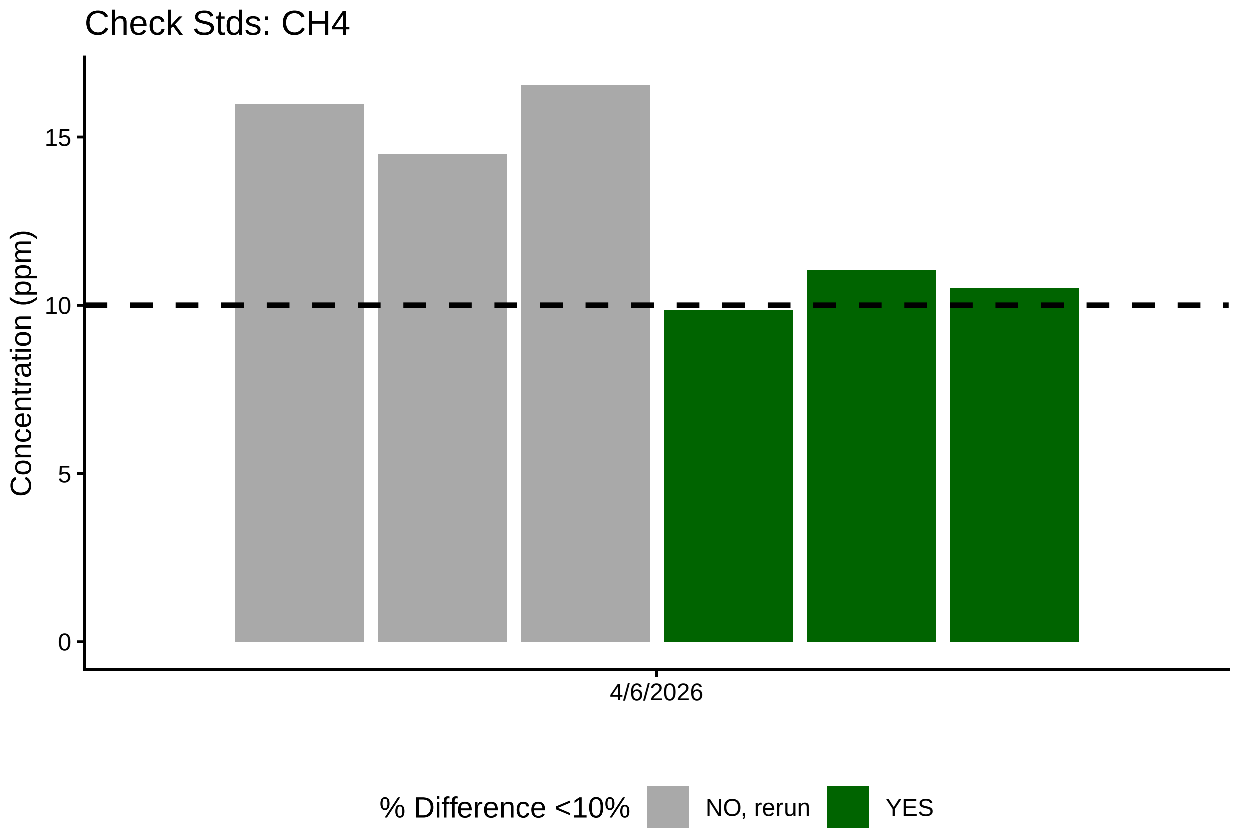
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



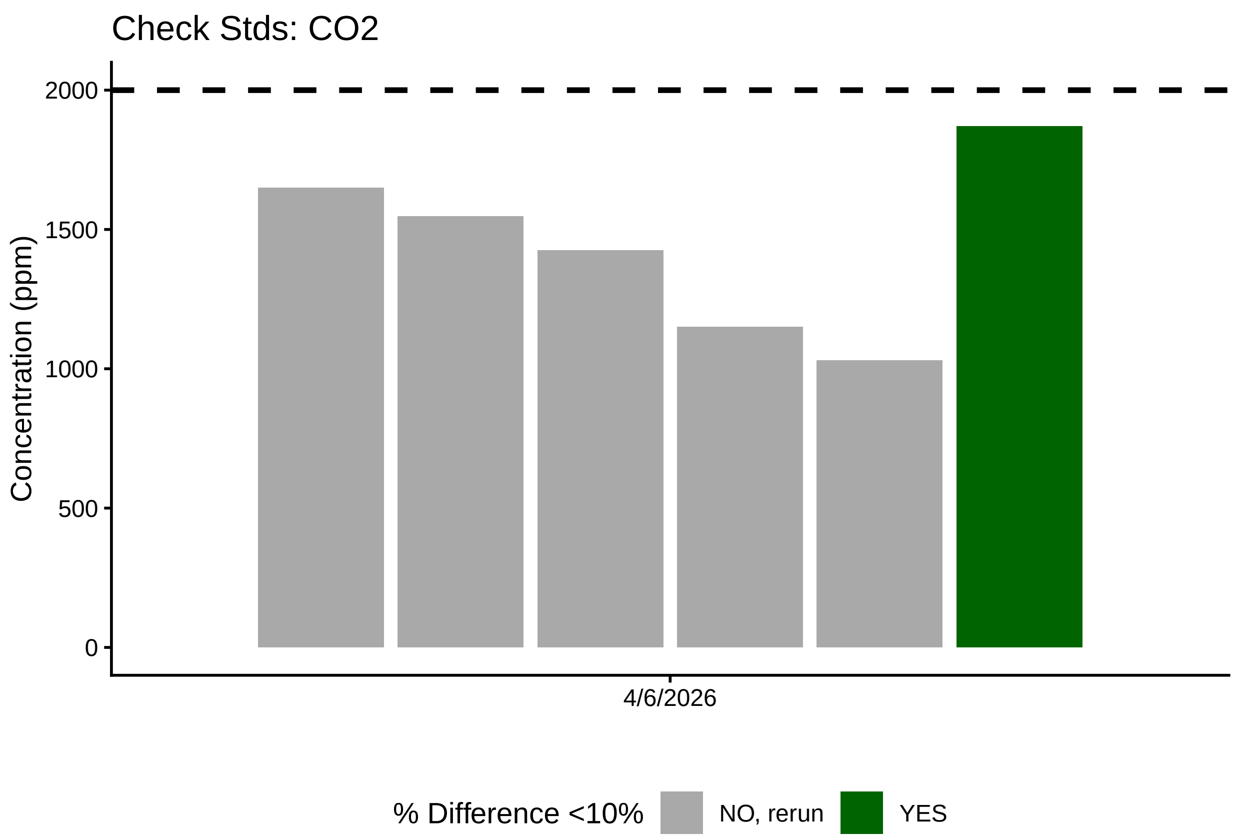
2.1 Now calculate the CH4 in ppm

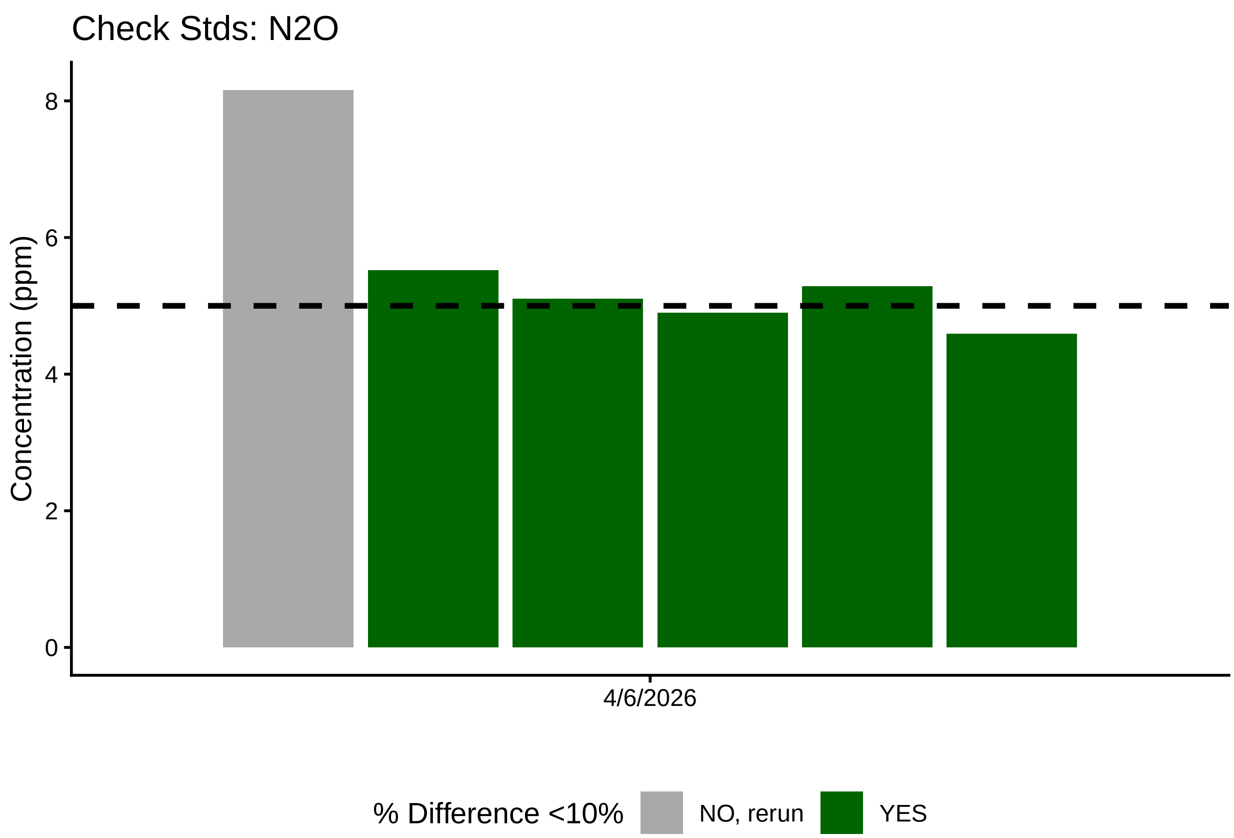
3 Aanlayze the Check Standards

```
## <60% of CH4 Check Standards are within range of expected concentration - REASSESS

## <60% of CO2 Check Standards are within range of expected concentration - REASSESS

## >60% of N2O Check Standards are within range of expected concentration - PROCEED
```



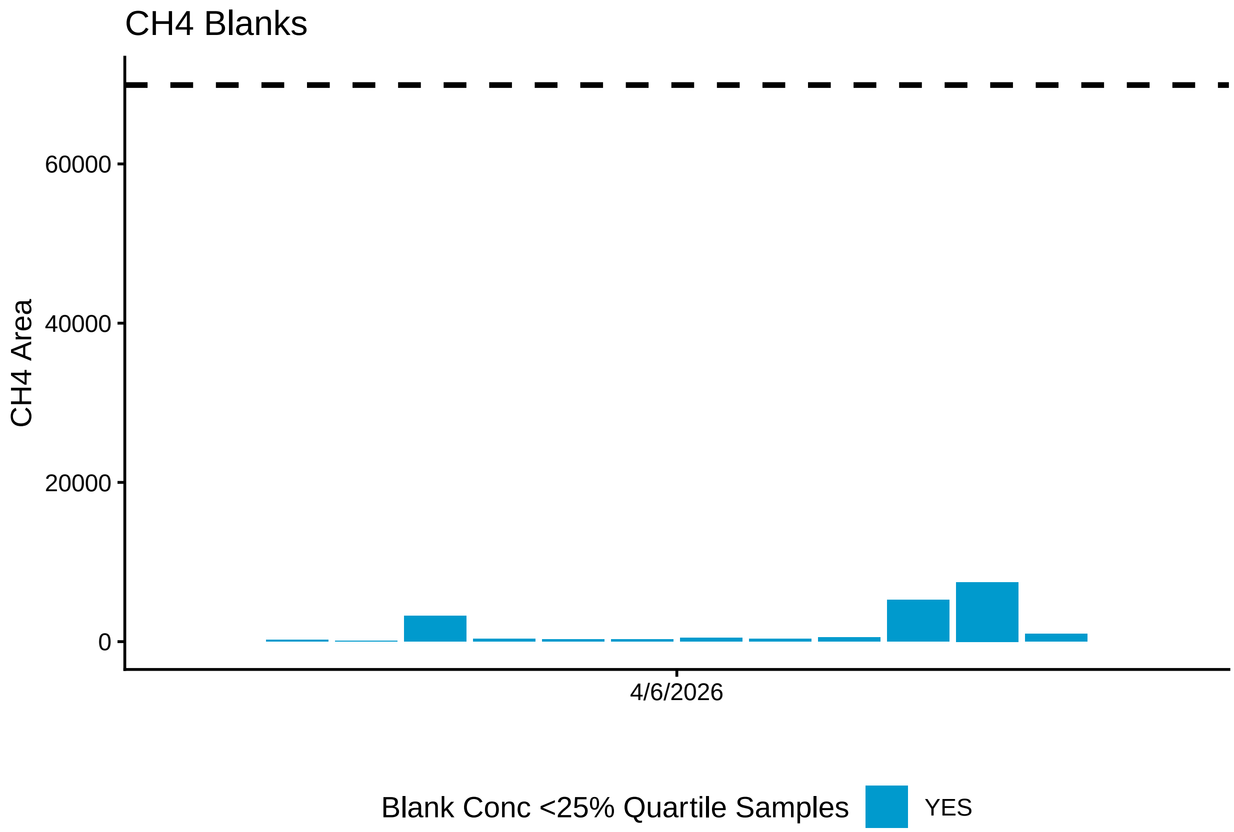
Check Stds: CH4
Concentration (ppm)
15
10
5
0
4/6/2026
% Difference <10%
NO, rerun
YES

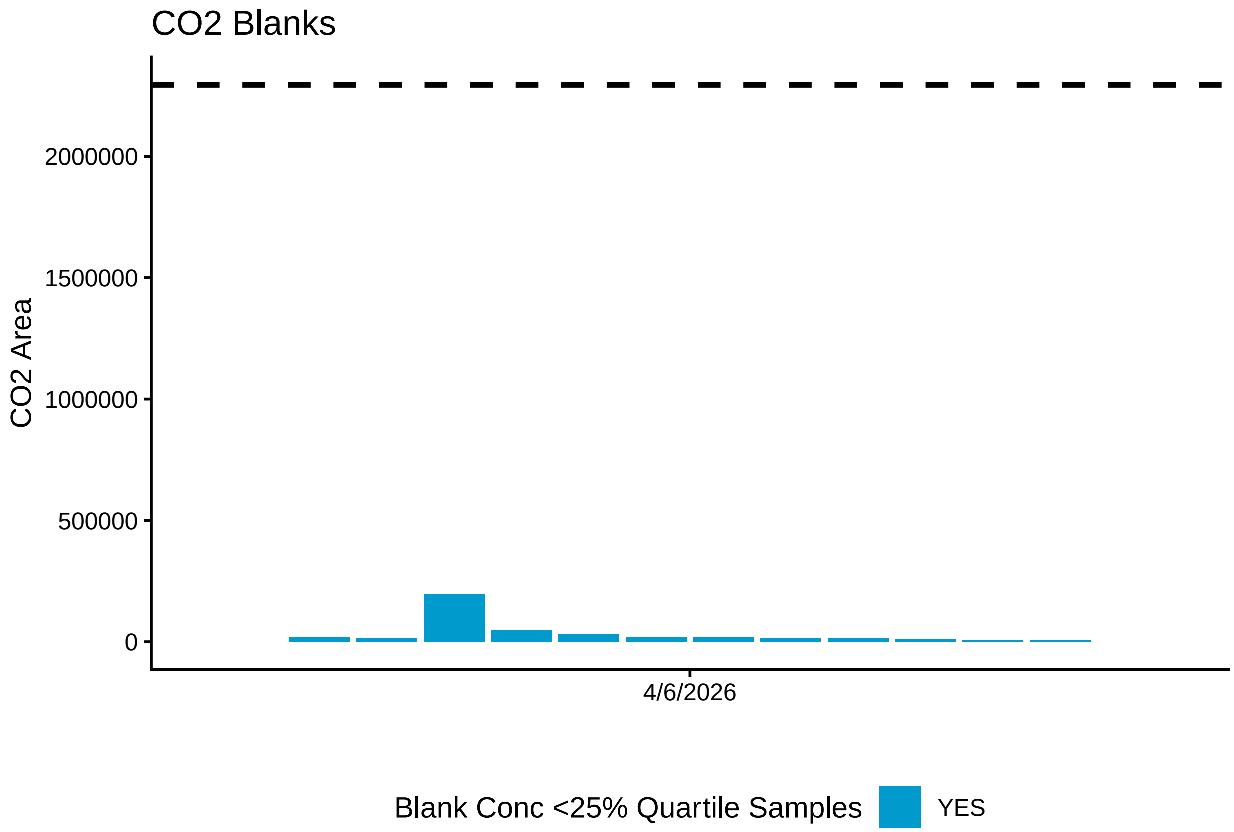

Check Stds: CO2
Concentration (ppm)
2000
1500
1000
500
0
4/6/2026
% Difference <10%
NO, rerun
YES



4 Analyze the Blanks

```
## Assess Blanks

## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED

## >60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED
```


CH4 Blanks
60000
40000
20000
0
CH4 Area
4/6/2026
Blank Conc <25% Quartile Samples
YES


CO2 Blanks
CO2 Area
2000000
1500000
1000000
500000
0
4/6/2026
Blank Conc <25% Quartile Samples
YES

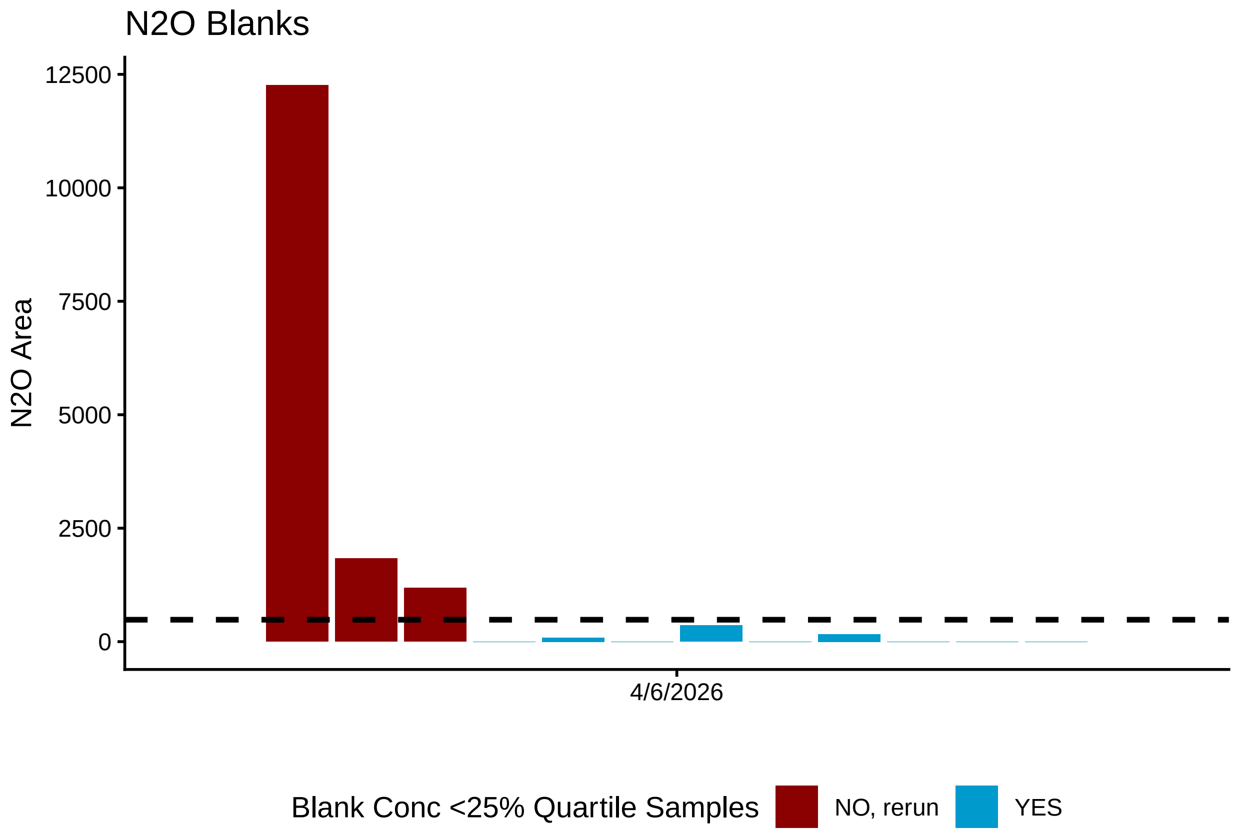
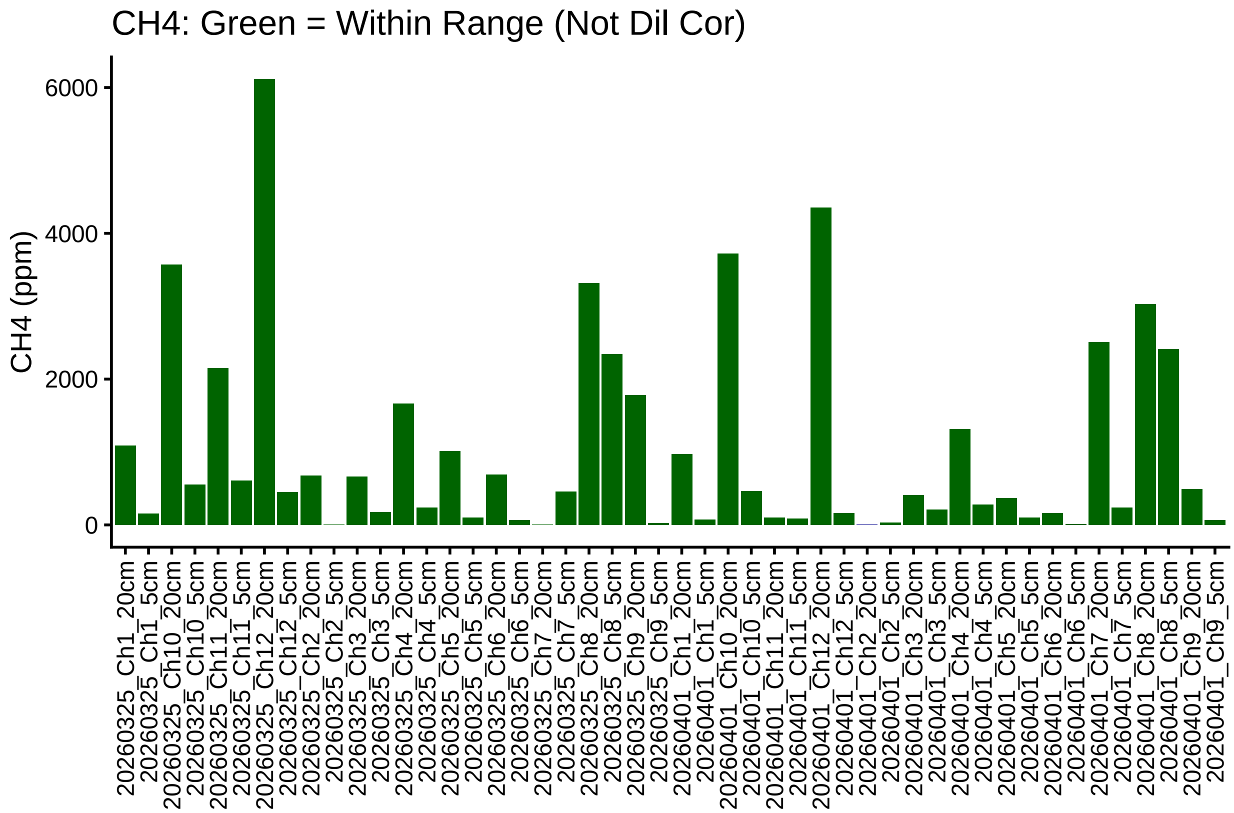
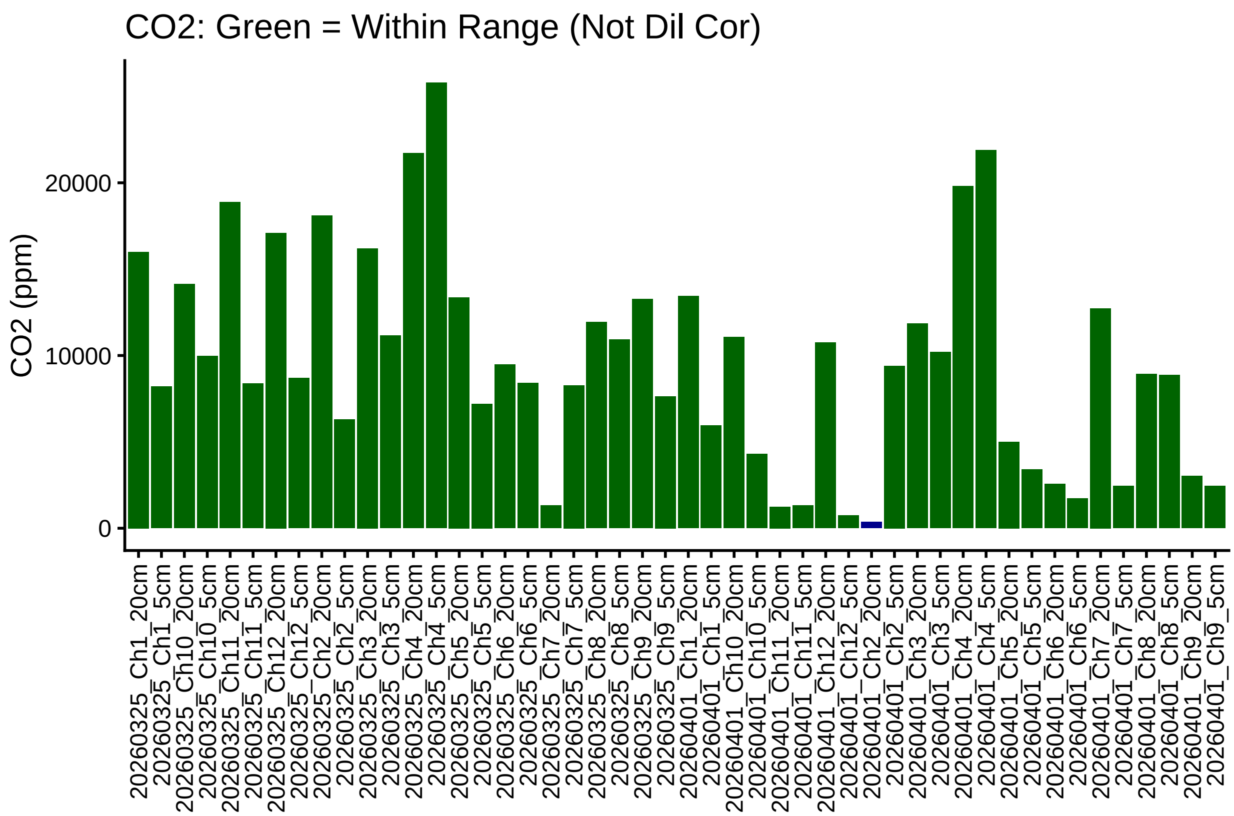


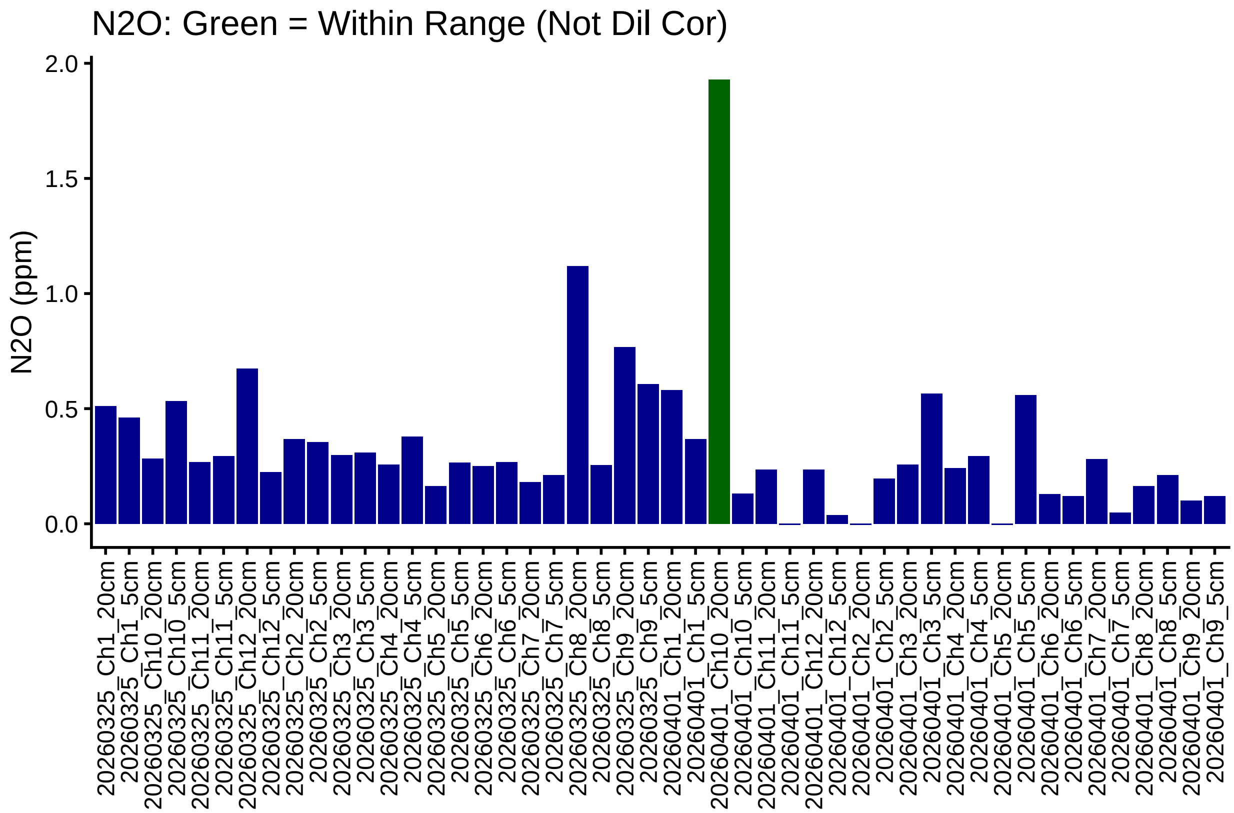
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	1623.000
CO2	33500.083
N2O	1324.583

5 Dilution correct samples and Check Range

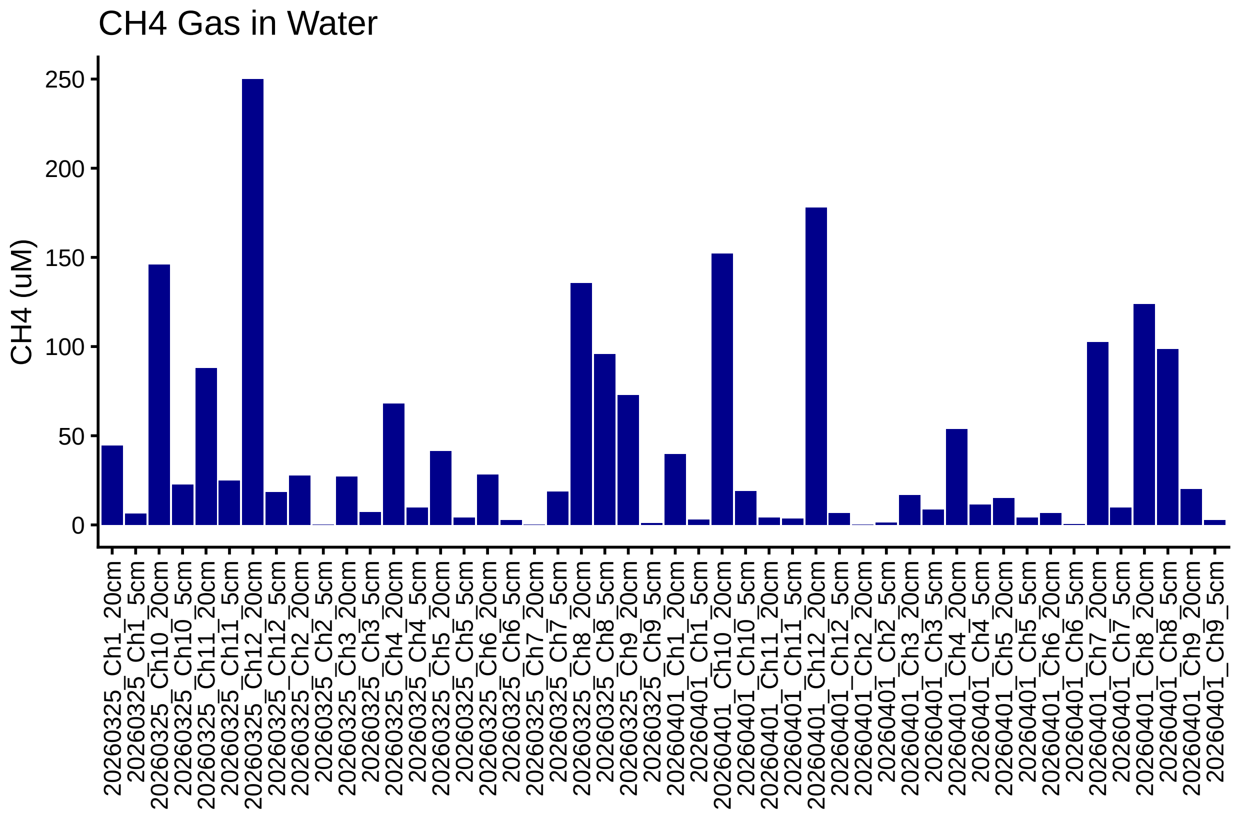


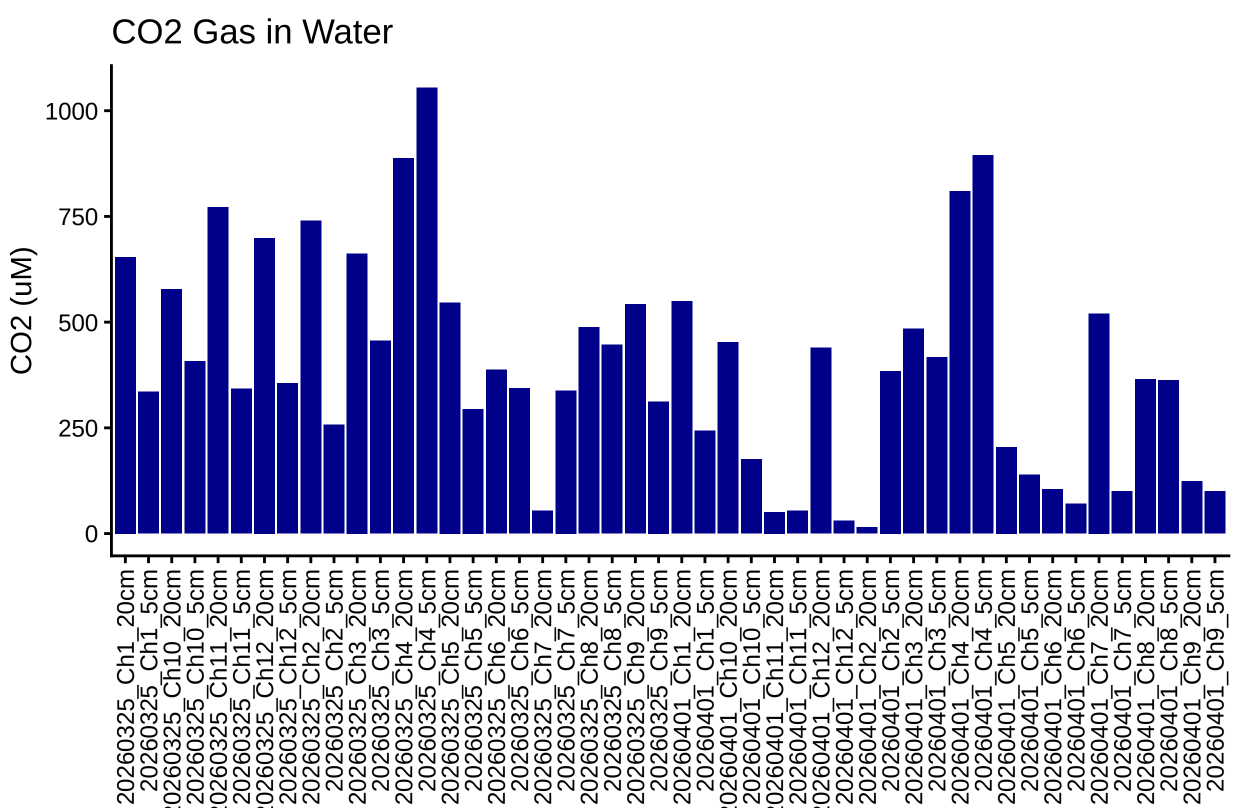


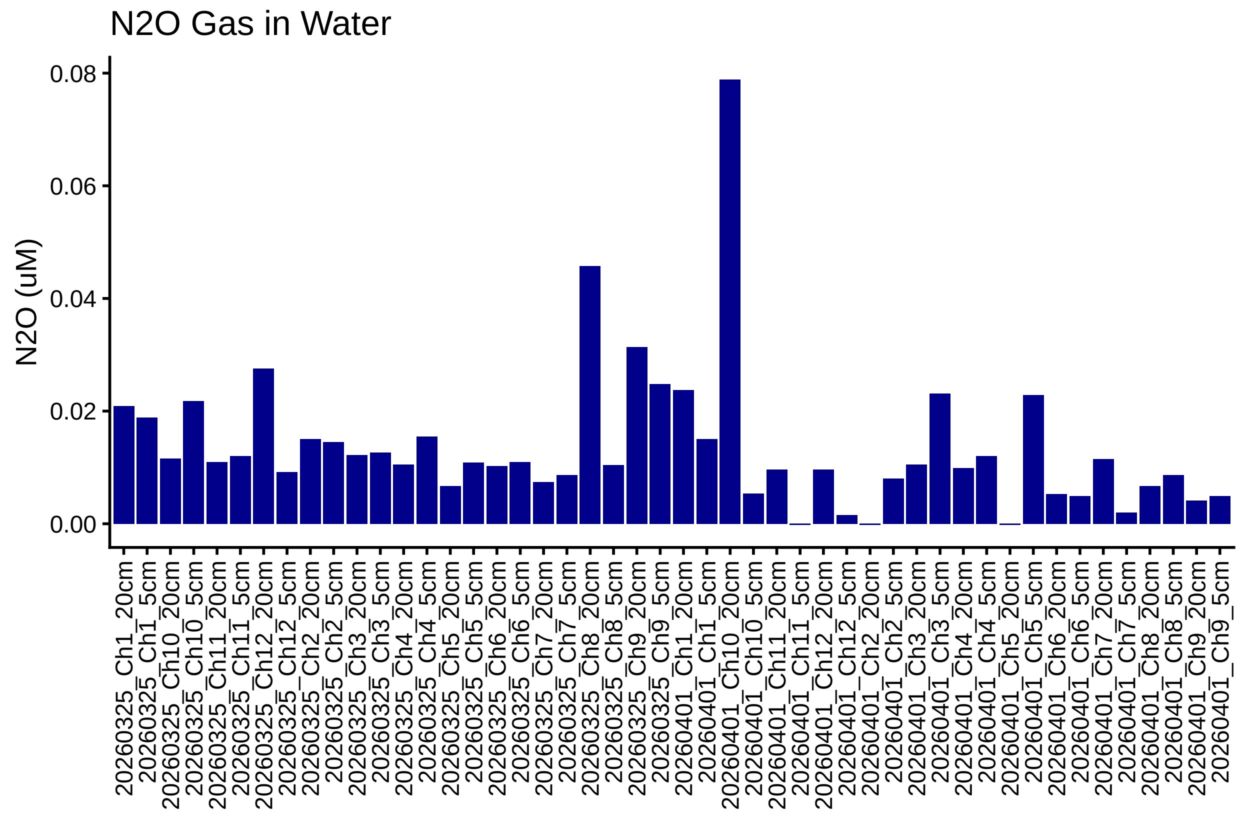


N2O: Green = Within Range (Not Dil Cor)
N2O (ppm)
2.0
1.5
1.0
0.5
0.0
20260325_Ch1_20cm
20260325_Ch1_5cm
20260325_Ch10_20cm
20260325_Ch10_5cm
20260325_Ch11_20cm
20260325_Ch11_5cm
20260325_Ch12_20cm
20260325_Ch12_5cm
20260325_Ch2_20cm
20260325_Ch2_5cm
20260325_Ch3_20cm
20260325_Ch3_5cm
20260325_Ch4_20cm
20260325_Ch4_5cm
20260325_Ch5_20cm
20260325_Ch5_5cm
20260325_Ch6_20cm
20260325_Ch6_5cm
20260325_Ch7_20cm
20260325_Ch7_5cm
20260325_Ch8_20cm
20260325_Ch8_5cm
20260325_Ch9_20cm
20260325_Ch9_5cm
20260401_Ch1_20cm
20260401_Ch1_5cm
20260401_Ch10_20cm
20260401_Ch10_5cm
20260401_Ch11_20cm
20260401_Ch11_5cm
20260401_Ch12_20cm
20260401_Ch12_5cm
20260401_Ch2_20cm
20260401_Ch2_5cm
20260401_Ch3_20cm
20260401_Ch3_5cm
20260401_Ch4_20cm
20260401_Ch4_5cm
20260401_Ch5_20cm
20260401_Ch5_5cm
20260401_Ch6_20cm
20260401_Ch6_5cm
20260401_Ch7_20cm
20260401_Ch7_5cm
20260401_Ch8_20cm
20260401_Ch8_5cm
20260401_Ch9_20cm
20260401_Ch9_5cm

6 Calculate gas in water







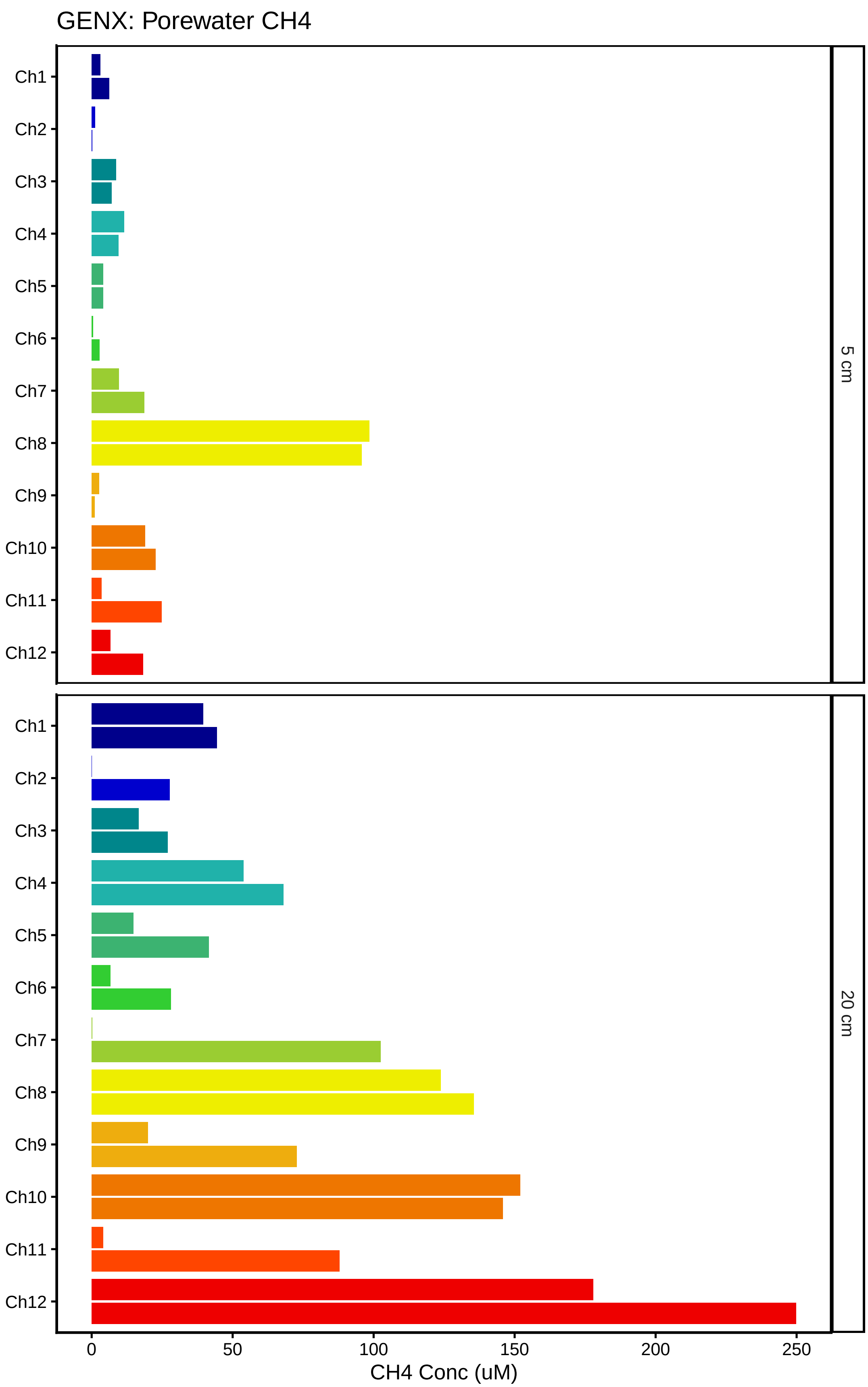
7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata

## Some sample IDs are missing from data.

##  [1] "Ch1-40"  "Ch1-80"  "Ch2-40"  "Ch2-80"  "Ch3-40"  "Ch3-80"  "Ch4-40"
##  [8] "Ch4-80"  "Ch5-40"  "Ch5-80"  "Ch6-40"  "Ch6-80"  "Ch7-40"  "Ch7-80"
## [15] "Ch8-40"  "Ch8-80"  "Ch9-40"  "Ch9-80"  "Ch10-40" "Ch10-80" "Ch11-40"
## [22] "Ch11-80" "Ch12-40" "Ch12-80"
```

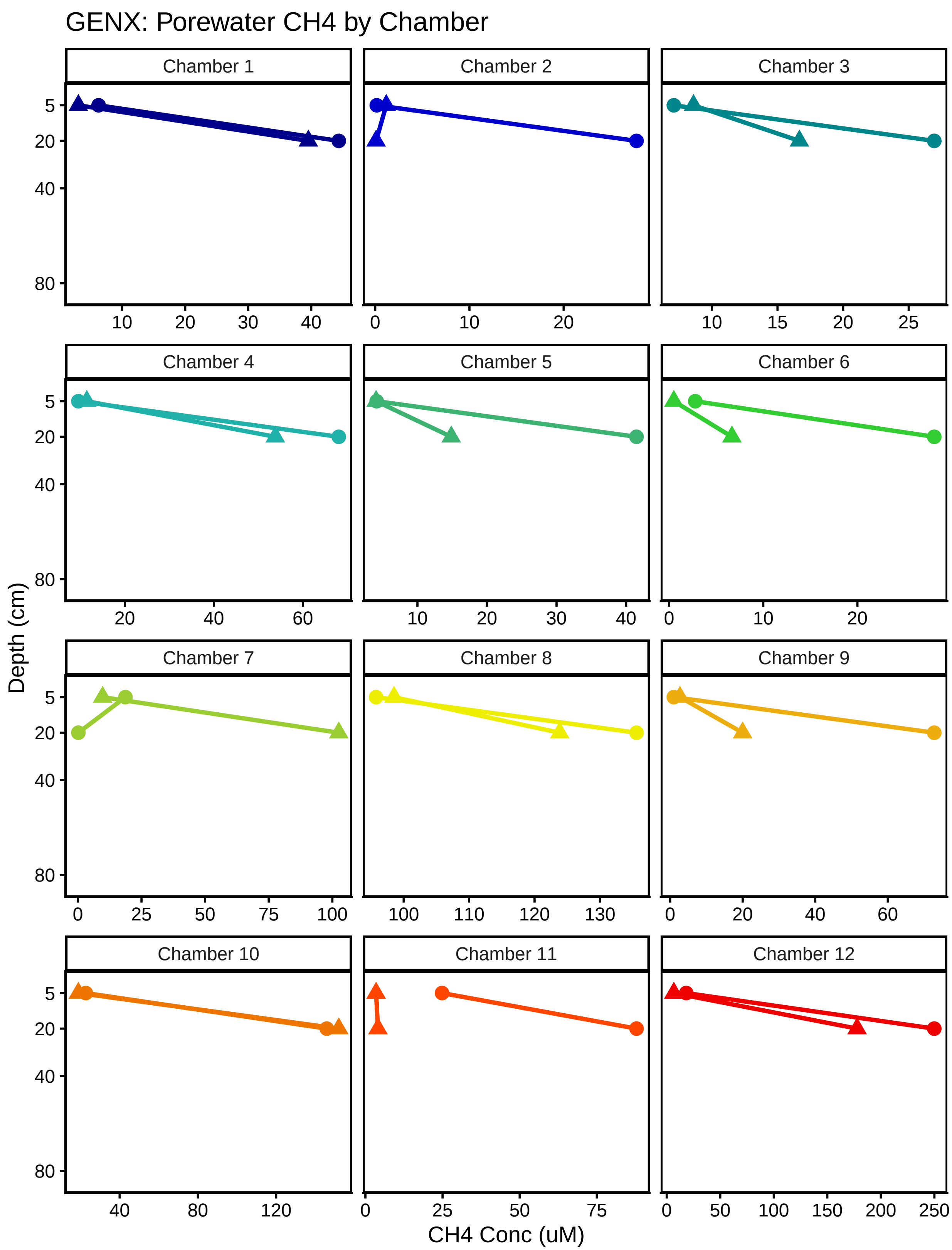
7.1 Organize Data

7.2 Visualize Data CH4

```
## Visualize Data
```

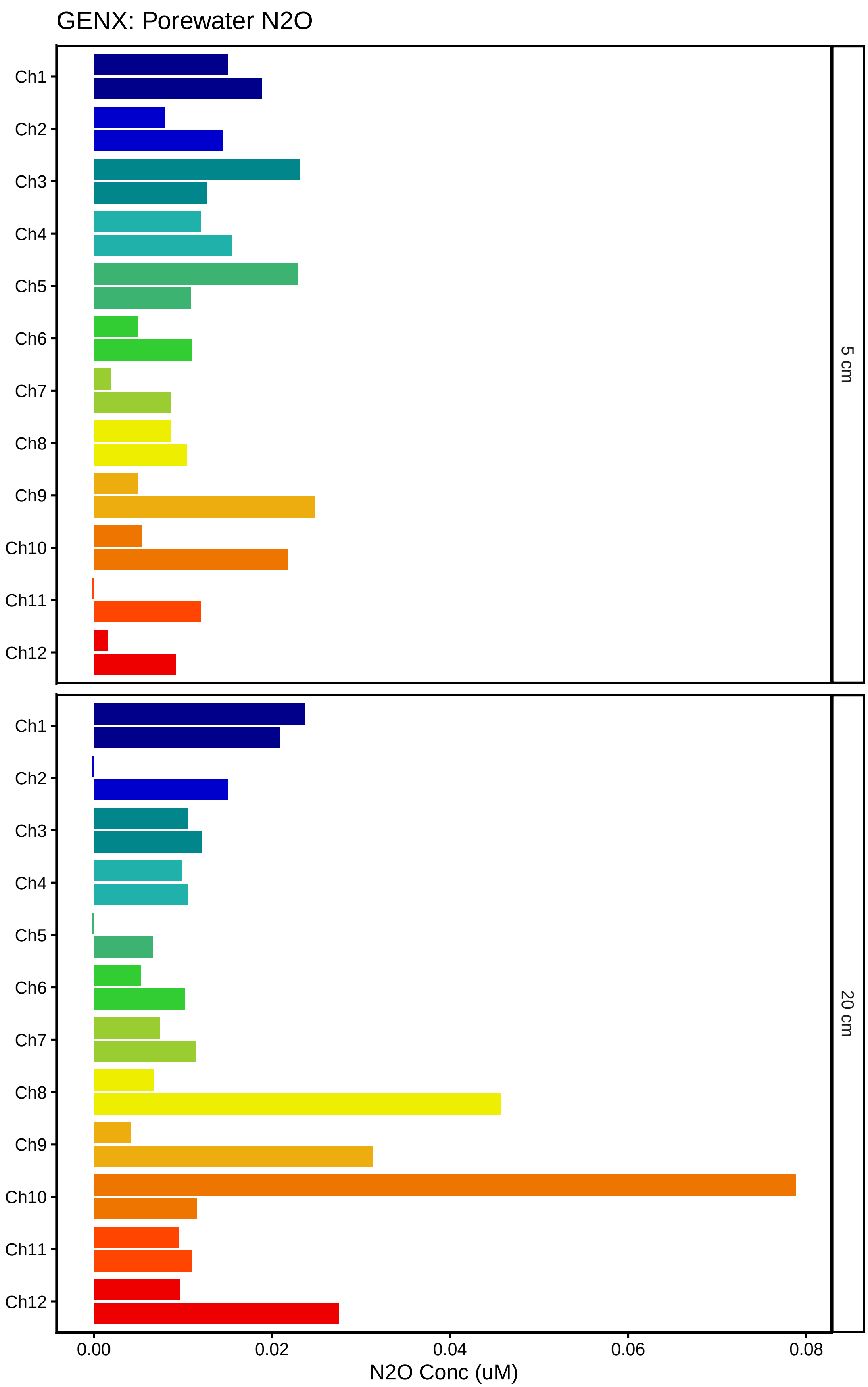


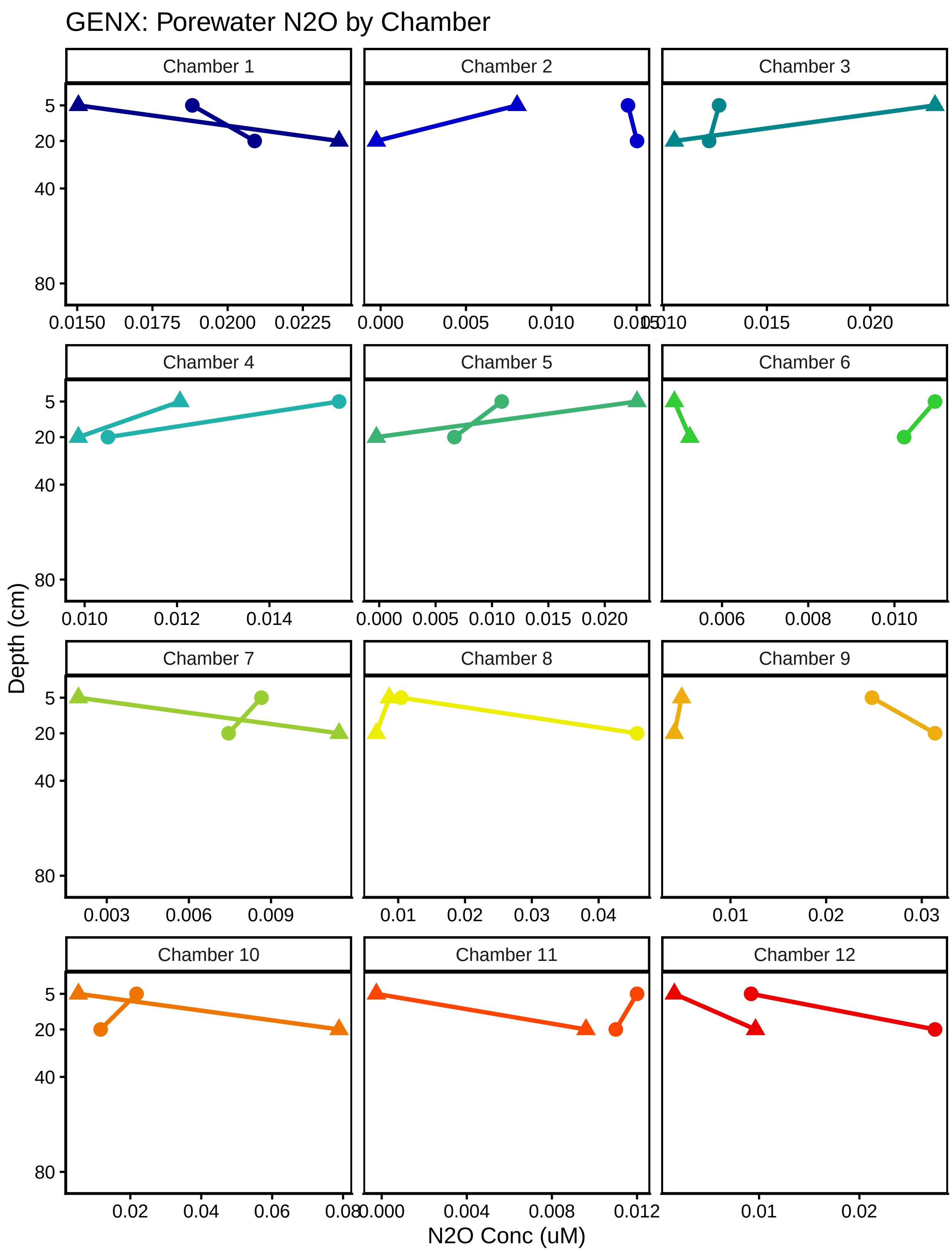
GENX: Porewater CH4
Ch1
Ch2
Ch3
Ch4
Ch5
Ch6
Ch7
Ch8
Ch9
Ch10
Ch11
Ch12
5 cm
20 cm
0
50
100
150
200
250
CH4 Conc (uM)



##Visualize Data N2O

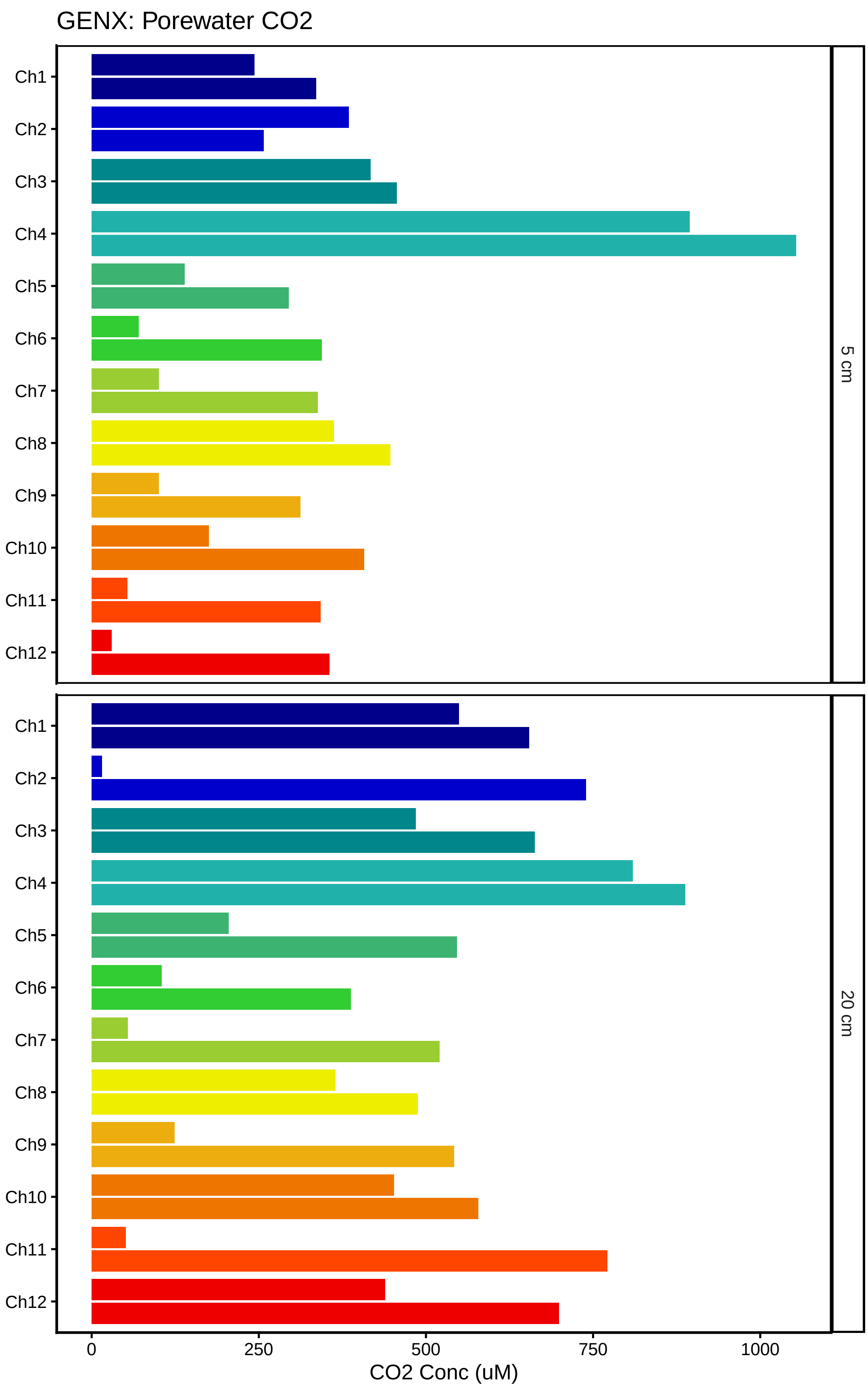
```
## Visualize Data
```

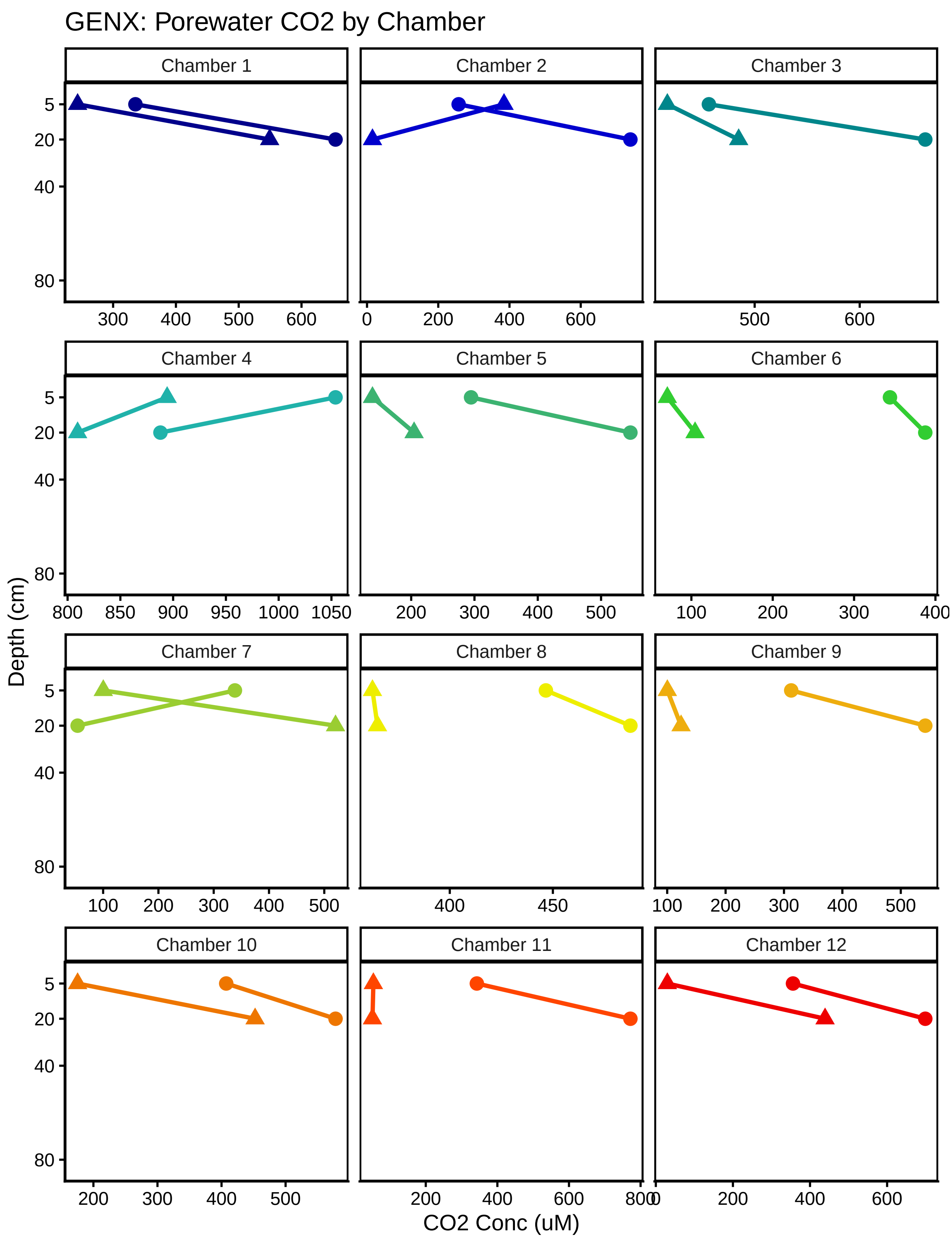

GENX: Porewater N2O
Ch1
Ch2
Ch3
Ch4
Ch5
Ch6
Ch7
Ch8
Ch9
Ch10
Ch11
Ch12
5 cm
20 cm
0.00
0.02
0.04
0.06
0.08
N2O Conc (uM)



##Visualize Data CO2

```
## Visualize Data
```


GENX: Porewater CO2
Ch1
Ch2
Ch3
Ch4
Ch5
Ch6
Ch7
Ch8
Ch9
Ch10
Ch11
Ch12
5 cm
20 cm
0
250
500
750
1000
CO2 Conc (uM)



##Export Data

#end